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When the Water Goes Marching in: Integrating Geosciences with Reservoir Management to Handle Water Breakthrough in a Mature, Highly Fractured, Carbonate Reservoir Field

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Abstract

Approximately 70% of the world's hydrocarbon reserves are contained in naturally fractured carbonate reservoirs. Despite their exceptional deliverability and favorable reservoir characteristics, managing these reservoir types poses distinct challenges, particularly when dealing with fluids of varying mobilities. Addressing these complexities necessitates: 1) an agile geoscience workflow capable of rapidly integrating and interpreting subsurface data, 2) adaptability in decision-making and field development strategies, including criteria for identifying drillable targets and optimizing well designs, and 3) comprehensive analysis of static and dynamic reservoir data to guide management and operational adjustments effectively.

A fictional, representative highly fractured, heterogeneous Middle East carbonate field was designed to characterize and model water pathways. The work focuses on the management of water during significant water breakthrough in a mature field, aiming to integrate geosciences and reservoir engineering to trace probable fluid flow pathways. Three critical datasets were identified: fracture characteristics, continuous bottomhole pressure logs for the wells, and seismic attributes to detail inter-well reservoir properties.

A fracture catalogue was developed to assess which fractures likely contribute to fluid transport. Water presence and its dynamic behavior was closely analyzed and followed up from existing well pressure and production data. Seismic attributes reveal reservoir discontinuities beyond well drainage volumes. Hypothesized flow paths, based on these tools, were validated against observed well performance.

As a result of the applied methodology, the most probable pathways for both oil and water movement are determined, leveraging dynamic data that can be continuously updated as new field information becomes available. This approach is valuable for identifying active fracture networks, critical for production optimization and the selection of new drillable targets. It supports decisions such as adjusting planned well trajectories across under-drained areas, managing production on existing wells, or modifying open producing intervals in selective completions. Furthermore, the methodology enables the hierarchization of fracture networks as fluid paths under current conditions, providing insights to guide infill drilling campaigns and improve the efficiency of the field production at different stages of its evolution.

This approach surpasses conventional reservoir analysis methods in both speed and reliability by integrating an extensive static and dynamic dataset with multidisciplinary expertise. Each map realization encourages collaborative discussions, leveraging specialized knowledge across geoscience, reservoir engineering, and production disciplines. The workflow is completed in significantly less time than traditional reservoir modelling, making it more agile and efficient. Additionally, it serves as a team-building process, promoting dialogue among subsurface, production, and field teams. Being heavily data-driven, this methodology reduces bias compared to conventional reservoir modelling, ensuring more objective insights and facilitating informed decision-making.

Introducing the Case Study

The current analysis is based on a recreated, representative fractured carbonate oilfield located in North Iraq. This region is home to significant carbonate oilfields, characterized by its complex geological structures and rich hydrocarbon resources. Fields are often marked by a series of fractures and faults which serve as pathways for hydrocarbon migration, enhancing their productivity. Oil in this area is in general medium to light gravity, with some exceptions. Porosity and permeability of the carbonate rocks are variable, influenced by the degree of fracturing and diagenetic processes. Deviated and horizontal drilling are in general the preferred strategy of increasing a well productivity by providing a higher reservoir exposure and the possibility to intersect more oil-bearing fault and fracture systems. Our case study, an oilfield located within the foothills of the Zagros Mountains, showcases a unique stratigraphy that includes a variety of carbonate formations, primarily composed of limestone and dolomite. Key characteristics of the field are listed below:

- Reservoir lithological facies: Limestone, dolomite
- Reservoir thickness: 200 m
- Reservoir Depth: 1300-1500 mTVDSS
- Fluid type: Oil, 27 API
- Trap type: Anticline, 3-way structural closure + Bounding Fault
- Wells drilled and on production: 5
- Well Type: Horizontal
- Reservoir Exposure length: along wellbore 500-1000 m
- Matrix Porosity: 4%
- Reservoir Permeability: Fractures: Up to 5 Darcy; Matrix: 5-15 millidarcy
- Initial Reservoir Pressure: 3060 psi

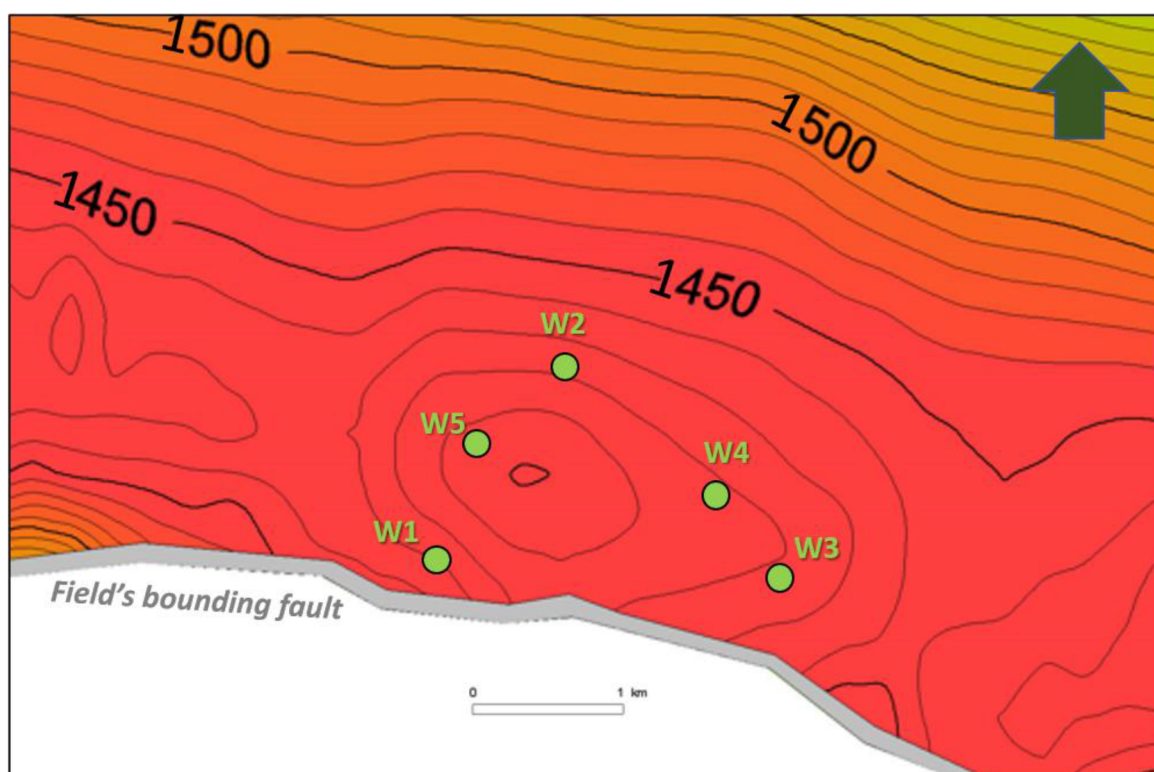


Figure 1—Depth Structural Map (mTVDSS) at Top or Reservoir for the field under evaluation, including well penetrations at this level.

Static Characterization of Fault and Fracture Elements

Seismic Interpretation

Experience has shown that in fracture carbonate fields, a focused analysis of fault and fracture networks during the seismic interpretation phase is crucial for accurate results. This process involves meticulous manual interpretation of reflectivity datasets, the application of geometric seismic attributes specifically designed to highlight these systems, and the utilization of relevant seismic datasets obtained through specialized processing techniques. Common geometric attributes that can typically be directly employed in seismic interpretation software include coherence, curvature, dip, azimuth, and semblance, among others. These attributes play a vital role in the understanding of the reservoir architecture, revealing the orientation, size, and shape of faulted and fractured zones. Figure 2 illustrates the fault and fracture networks highlighted from a coherence attribute calculation. Coherence evaluates the correlation of the seismic waveforms within a window. Areas of low coherence highlight chaotic patterns and discontinuity zones. From this attribute calculation, it is observed that Wells 1 and 2 are strategically located within a prominent fault corridor, while Well 3 is situated in a secondary corridor that, although less distinct in the reflectivity data, becomes more apparent through the application of seismic attributes. Conversely, Wells 4 and 5 are positioned in areas where discontinuities are either not visible or clearly defined. It is important to recognize that a comprehensive understanding and identification of these fault and fracture systems may not solely rely on seismic data, especially in instances involving shallow reservoirs or complex geological formations, which can lead to a degradation in seismic imaging quality. Observations of mud losses during drilling operations often correlate directly with the presence of faults and fractured zones. Analyzing the mud loss data reveals that both Well 1 and Well 2 experienced severe-to-total mud losses, consistent with their locations within the main fault system identified on the seismic data. Well 3 also experienced moderate mud losses, aligning with its placement in a secondary fault corridor. Interestingly, while Well 4 did not encounter any mud

losses, Well 5 experienced moderate-to-severe losses, suggesting it may be situated within a damage zone associated with the nearby mapped main fault corridor. This information is critical for understanding the extent of disturbances caused by the main fault system, even if the well appears to be located outside of it. For production purposes, it is particularly noteworthy that Well 5 is anticipated to be part of the fluid pathways similar to Wells 1 and 2, indicating a more direct connection with these wells. This insight underscores the importance of thorough seismic interpretation and mud loss analysis in optimizing production strategies in fractured carbonate fields.

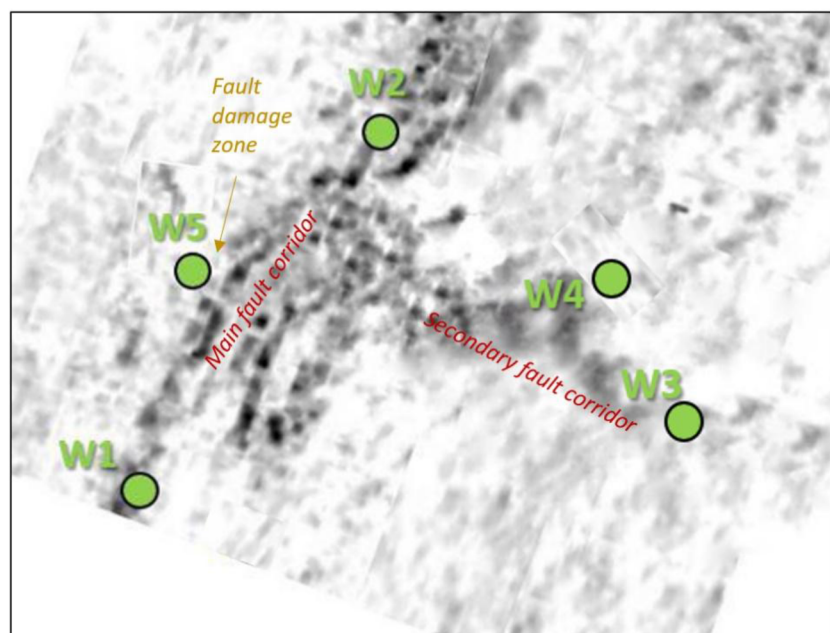


Figure 2—Seismic coherency map at top of reservoir for the area under evaluation, including well intersections.

Log Evaluation and Analogues of Flow-relevant Geo-parameters

Fluid pathways in structurally influenced carbonates can be the result of multi-fold combinations between inherent depositional facies architecture, diagenetic variability and a variety of fault and fracture configurations. This translates into a diverse flow ecosystem with pronounced lateral and vertical variability which can be observed in the showcased reservoir of this study. The wireline log responses, well performances and bottomhole flowing pressure signatures (FBHP) of the represented horizontal and slanted wells suggested distinctive differences in drainage volumes and character in pathway mobility of the investigated reservoir fluids oil and water. Seismic attribute characterization proposed the presence of seismic-scale deformation features that were considered as drilling targets. The prospects differed in terms of width, orientation and depth extend of reflector disturbance and the relative distance of the wellbores to the different degrees of mapped deformation features.

Early dynamic indications during drilling of the wells like mud losses, changes in ROP, abrupt resistivity changes and gas readings confirmed congruency between the mapped fault zones and their active capacity for fluid flow. Image logs revealed a variety of fracture categories across the seismic-scale features and the presence of background fractures in relative distance to fault damage zones with variable degrees of systematic orientations. Mud-losses and subsequent flow rates appeared to be influenced by differences in fracture typology; hence differentiation into fracture categories like bed-bound fractures, diagenetically altered vuggy fractures of variable apertures, fracture corridors, fracture clusters and fault damage zones with associated brecciation and outsized pores were considered as drivers for distinctive differences in cumulative rates and inflow profiles. Periodic Production Logging Tool (PLT) surveys validated the distinguished dynamic hydraulic capacity of the different fracture categories according to their inflow

profile. The iterative development of a catalogue representing the dynamic ranges per geo-parameter (e.g. fracture corridors vs. vuggy fracture cluster) was attempted for assessing and classifying the degree of flow partitioning between each geo-parameter (see Figure 3).

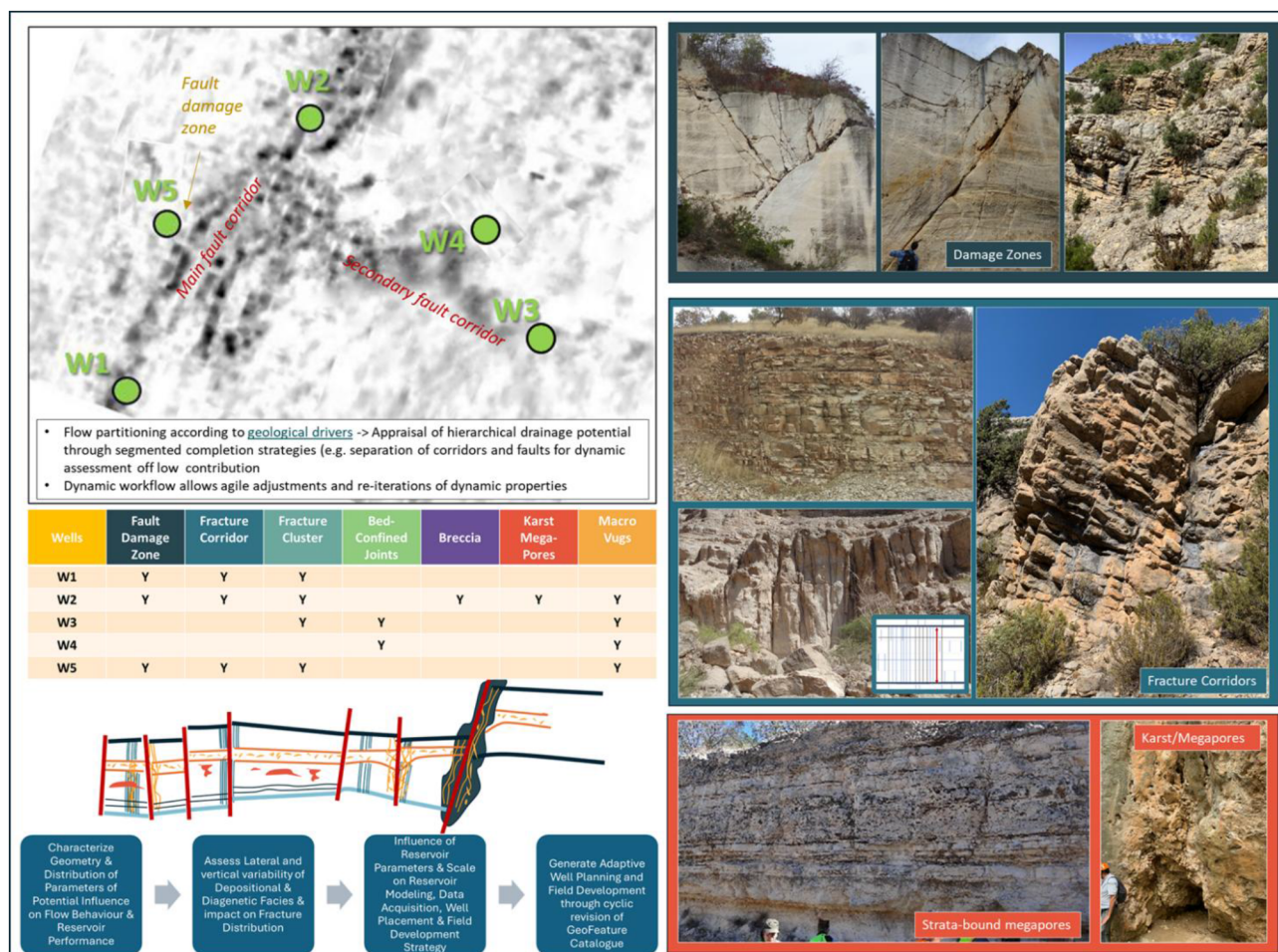


Figure 3—Determining Geo-Parameters and their spatial distribution as flow pathways. Outcrop analogues offered conceptual inspiration to evaluate dimensional characteristics and litho-mechanic drivers for fault-zone architecture and fracture distribution that were identified on image logs and seismic attributes. The evaluation of hierarchical flow regimes responsible for the observed pressure signatures and water cuts was designed as a dynamic assessment process to capture the constitution and interaction of drainage bodies related to the identified geo-parameters.

Conceptual flow scenarios were established according to the static (mainly based on image logs) and dynamic calibration points along the well trajectories in order to track the lateral and vertical continuity of pathways and permeability contrasts and to study their potential interaction between different wells. Dimensional characteristics were directed by seismic attribute analysis and inspired by outcrop analogs to refine the conceptual realization of flow scenarios and to rank the likelihood of water entry points along the well trajectories (see Figure 4).

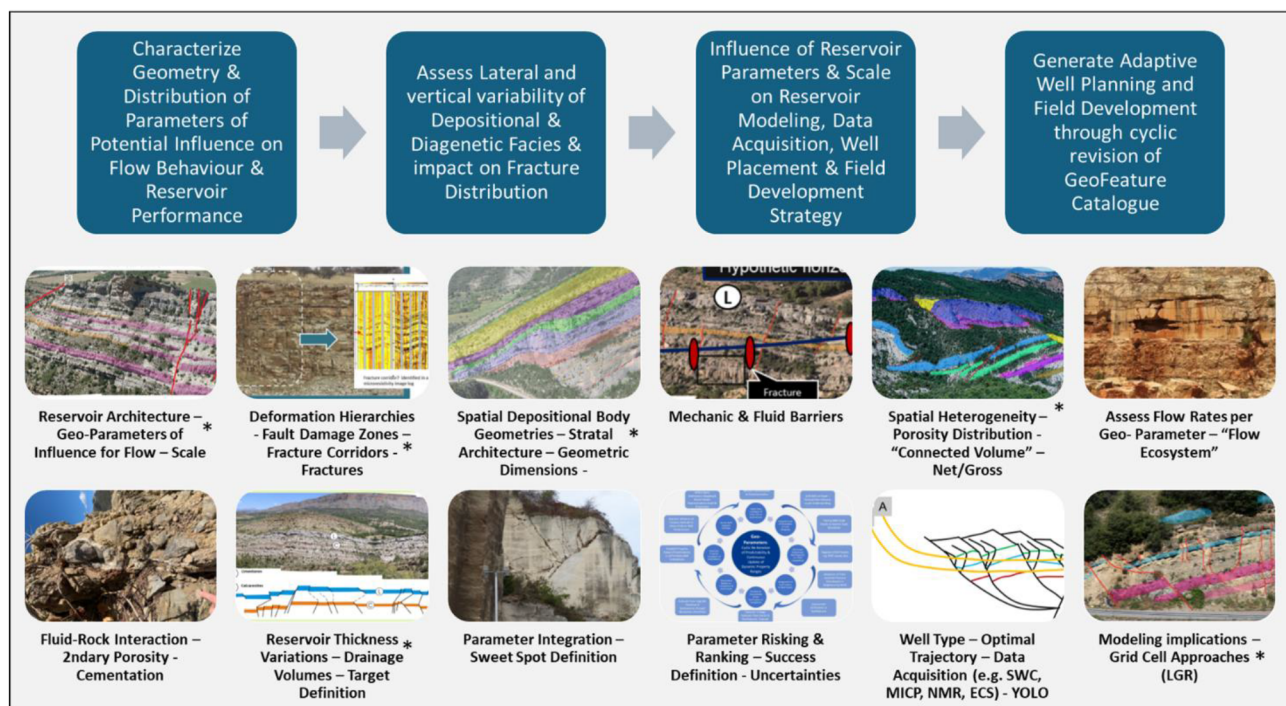


Figure 4—The inherent complexity of the fluid pathway architecture of the reservoir showcase inspired the design of a continuous reservoir characterization based on the multiple simultaneous operational activities (e.g. drilling, workovers, water cut mitigations). Continuous integration between the static catalogue of geo-parameters and their dynamic signatures enables continuous improvement in understanding reservoir dynamics through space and time. Increased adaptability to events such as water breakthroughs allows implementation of new dynamic data calibration and field development adjustments. Selected visual elements presented in this diagram are adopted from Indivo GeoConsult field guides with contributions from Jon Gutmanis, Lluís Ardevol I Oro and Valenti Turu. Figure modified after Olmeda & Funk (2024).

The streamlined data feedback analysis between the reservoir surveillance campaign (see Table 1) and the static geo-parameter characterization was the foundation for the establishment of an agile and proactive reservoir management. (see Figure 5).

Table 1—Surveillance Campaign Recommended for Wells in Naturally Fractured Reservoirs with Active Bottom Aquifer

Category	Data Type / Tool	Purpose	Timing / Frequency
Static characterization	Image Logs (FMI, UBI)	Identify open fractures, orientation, density	All wells
	Core Acquisition + Analysis	Fracture properties, matrix porosity/permeability, capillary pressure	Selected wells
Post-Completion / Production	PLT / Array PLT	Inflow profiling, water entry detection	Baseline post-startup, then contingent
	Wellhead Surveillance (WHP, Rate, GOR, WC, Temp)	Performance and water cut monitoring	Hourly to daily (SCADA)
Pressure Monitoring	Permanent Downhole Gauges (PDG)	Reservoir depletion, aquifer movement	Continuous
	Interference / Pulse Tests	Fracture connectivity and permeability tensor	Periodically (after new well drilling)
	MDT / RFT	Pressure and mobility profiling	In pilot/infill wells
PVT & Fluids	PVT Sampling and Study	Bubble point, viscosity, Bo, Rs, compressibility	Selected wells at initial conditions
Water Management	Tracer Tests (optional)	Identify water source (aquifer vs fractures)	If water source uncertain
	RST / Pulsed Neutron (cased hole saturation logs)	Monitor water saturation trends	Every 1–2 years

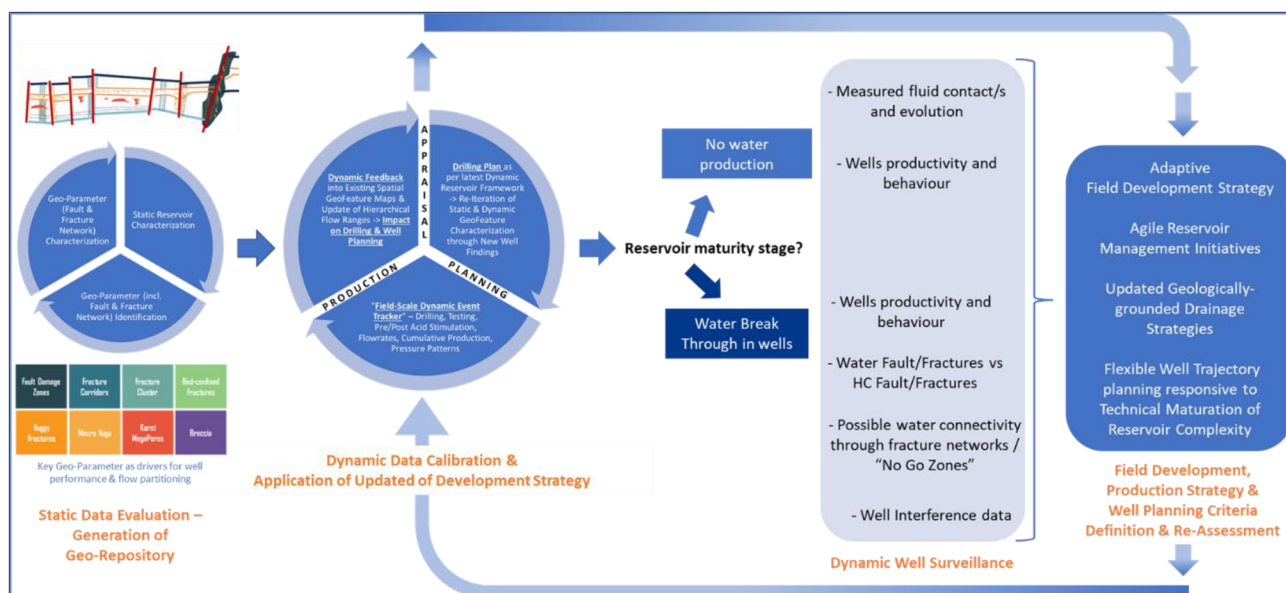


Figure 5—Circular Assessment Workflow: Revisiting dynamic calibration of key Geo-Parameters & Impact on Development Strategy & Well Planning Criteria.

The following chapter investigates the dynamic evolution of selected wells and their specific bottomhole flowing pressure (FBHP) profiles in response to water encroachment. Their varied response resembles their different repository in flow-relevant geo-parameters and their permeability contrasts, their distance to primary and secondary faults, general fault spacing and width of damage zone drainage, presence of fracture corridors, matrix potential and relative distance to initial water-oil-contact (see Figure 6).

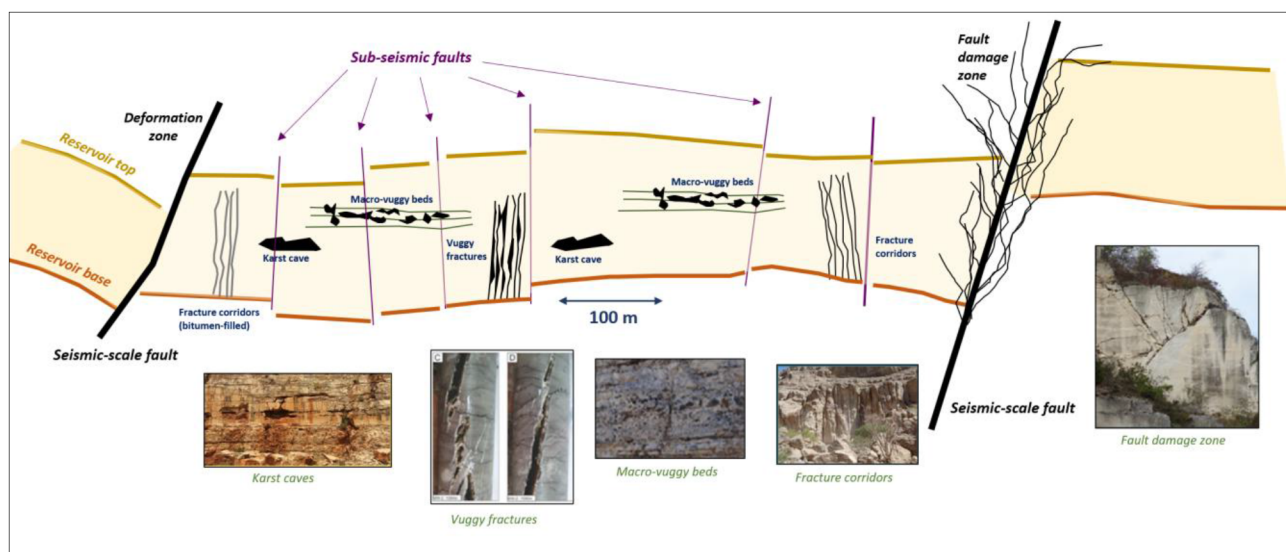


Figure 6—Schematic distribution of Geo-Parameters (as listed on Fig. 3) introducing lateral and vertical contrasts in permeability. Complex reservoir architecture in structurally controlled carbonate reservoirs often requires re-learning of reservoir understanding and adjustment of conceptual geological models and flow path assessments through space and time. Well planning and adaptable field development strategies offer agile data understanding and implementation of new reservoir learnings in production and well trajectory iterations, such as targeting or avoiding specific features such as fault damage zones. Figure modified after Olmeda & Funk (2024).

Dynamic Evolution of the Field & Reservoir Management

This section analyzes the reservoir's dynamic behavior from initial production through the onset of water breakthrough, focusing on five horizontal and slanted wells selectively completed in a thick, naturally

fractured carbonate reservoir (NFR). These wells are equipped with electrical submersible pumps (ESPs) and downhole pressure gauges installed both upstream (intake pressure) and downstream (discharge pressure) of the pump, in a setup shown in Figure 7.

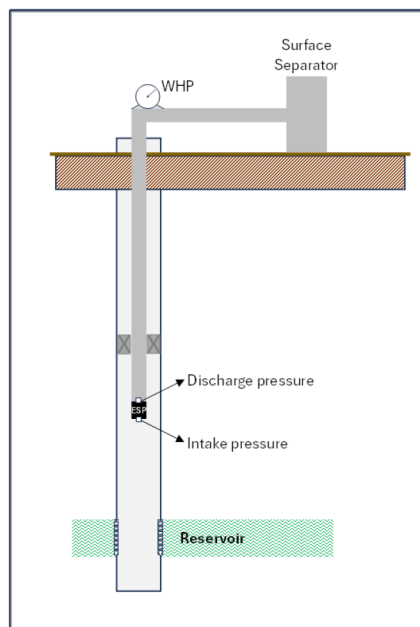


Figure 7—Wells discussed in this paper are completed with ESP Equipped with Downhole Gauges in a Setup Similar to the Scheme shown

For the purposes of this discussion, the intake pressure, measured between the pump and the producing intervals, is particularly relevant and will be referred to interchangeably as intake pressure or bottomhole pressure (BHP). Special attention is given to trends in BHP prior to water breakthrough, with the aim of identifying patterns that may serve as early indicators of water arrival. Recognizing these trends enables proactive reservoir management, allowing operators to implement timely mitigation strategies that delay water encroachment and reduce associated oil production losses.

In such reservoirs, characterized by natural fracturing, horizontal well architecture, undersaturated 27° API oil, and strong bottom aquifer support, continuous surveillance is critical for proactive reservoir management. Key parameters to monitor include high-resolution FBHP and reservoir pressure measurements, which are essential for understanding depletion behavior and tracking aquifer advance. Continuous surface data acquisition, encompassing production rates, water-cut, gas-oil ratio (GOR), and wellhead pressure, supports the early identification of inflow anomalies and water onset. Periodic Production Logging Tool (PLT) surveys, particularly using array or multiphase tools, provide inflow profiles along the lateral which are needed to localize water entry points and well segments with no inflow or low PI. Collectively, this suite of surveillance activities enables operators to better characterize fracture-matrix interactions, fine-tune drawdown strategies, and manage water production risks, thereby sustaining well performance and maximizing recovery.

A summary of the surveillance campaign carried out is provided in the Table below:

On its initial stage, the field produces highly undersaturated dry oil. Wells W1 and W2 are wells with high productivity, observing high permeabilities which are associated to their placement in a main fault corridor. Even though all wells are in communication in the field, the data indicates that the communication between these two wells is faster than the field's overall, confirming their placement into the same faulted cluster. Wells 3 and 5 rank second in terms of production contribution, with a relatively stable pressure behavior under controlled production conditions. Well 4, on the other hand, shows less contribution to the

overall production than all previous wells with a steeper pressure decline, indicating poorer permeability and reservoir connectivity to the dominating oil pathways. The reservoir has a strong bottom aquifer that partially replace produced fluids.

Regarding production behavior, all wells show different patterns depending on the geometrical set-up of the well, fractures and the aquifer.

Well W1

Water breakthrough in well W1 exhibits typical signs of bottom aquifer influx through high-conductivity fractures. This behavior is reflected in the shape of the bottomhole flowing pressure (BHP) curve. A previous study demonstrated that the vertical velocity of water approaching the well from a bottom aquifer correlates with an increase in the slope of the BHP curve, using a proxy known as The Water Towers. To understand the well's behavior in the context of its completion and the aquifer characteristics, close attention must be paid to the development of water-cut. The pressure performance of this well is shown in Figure 7 below.

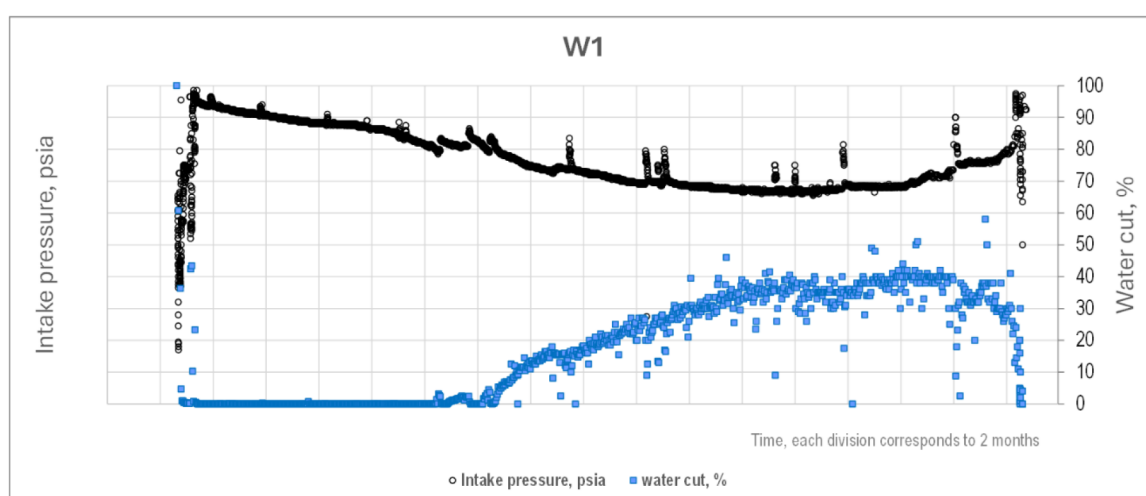


Figure 8—Bottomhole pressure and water cut of well W1

Figure 9 presents two plots, both displaying the bottomhole pressure. The upper plot shows bottomhole pressure alongside the liquid production rate, while the lower plot pairs it with the water cut. Three distinct time intervals are highlighted using colored panels, yellow, blue, and green, to facilitate interpretation of key changes in wells behavior.

Panel 1 highlights a shift in the trend of the well's flowing bottomhole pressure. A decline in pressure despite constant liquid offtake suggests water encroachment from the underlying aquifer. Toward the end of this period, a slight increase in water production is observed. This surge is mitigated by reducing the liquid rate, which subsequently lowers the bottomhole pressure, indicating a gravity-controlled system. Notably, the change in the slope of the bottomhole pressure curve provides an early warning that lasts for approximately three months.

Panel 2 shows the results of a deliberate reduction in liquid rate, executed in two semi-swinging stages. A moderate and temporary decrease, cutting the rate roughly in half, helped manage water production. The final reduction to 500 b/d successfully halted water-cut for approximately 15 days without causing further decline in reservoir pressure.

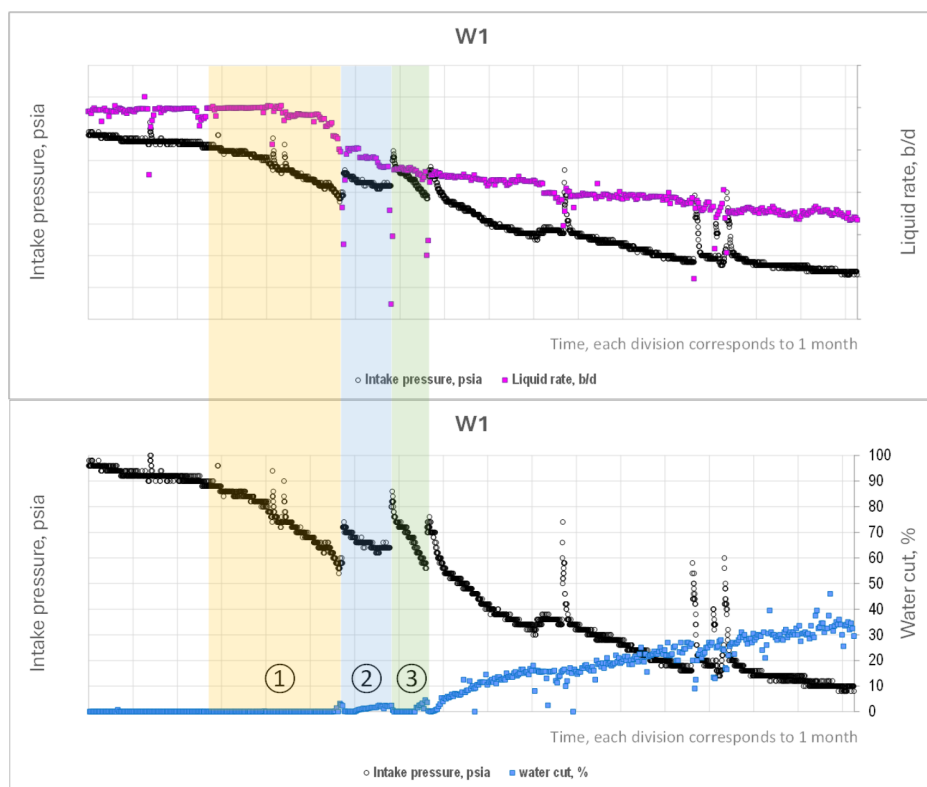


Figure 9—Detailed View of the Bottomhole Pressure, Liquid Rate and Water-Cut Behavior of Well W1

Panel 3 presents a final attempt to delay water production through another halving of the liquid rate. This measure temporarily mitigated water-cut for about one week. After this point, no further efforts were made to control water production via adjustments to the bottomhole flowing pressure.

Well W2

Well W2 is another producer located in the same structural corridor as W1 but positioned at a higher elevation. It exhibits a similar pressure trend to that of W1; however, the pressure drop is more pronounced due to the greater vertical distance between the well trajectory and the initial water-oil contact (WOC), compared to W1. Figure 10 illustrates the pressure behavior of W2.

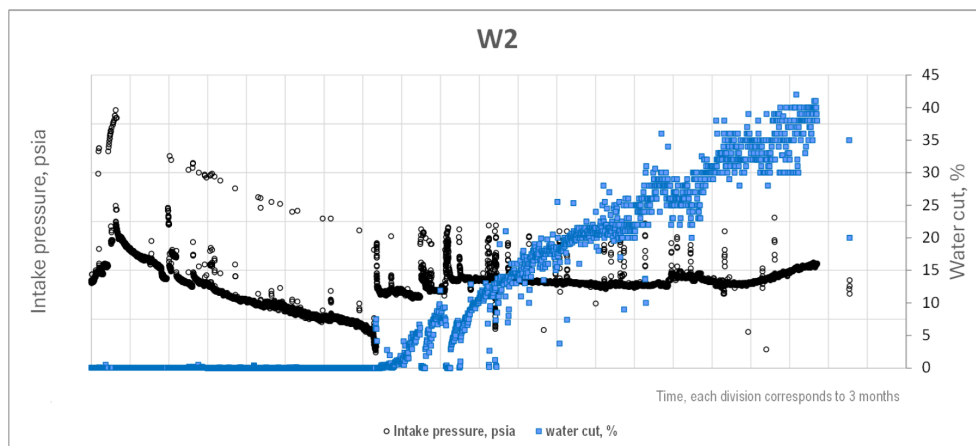


Figure 10—Bottomhole pressure and water-cut of well W2

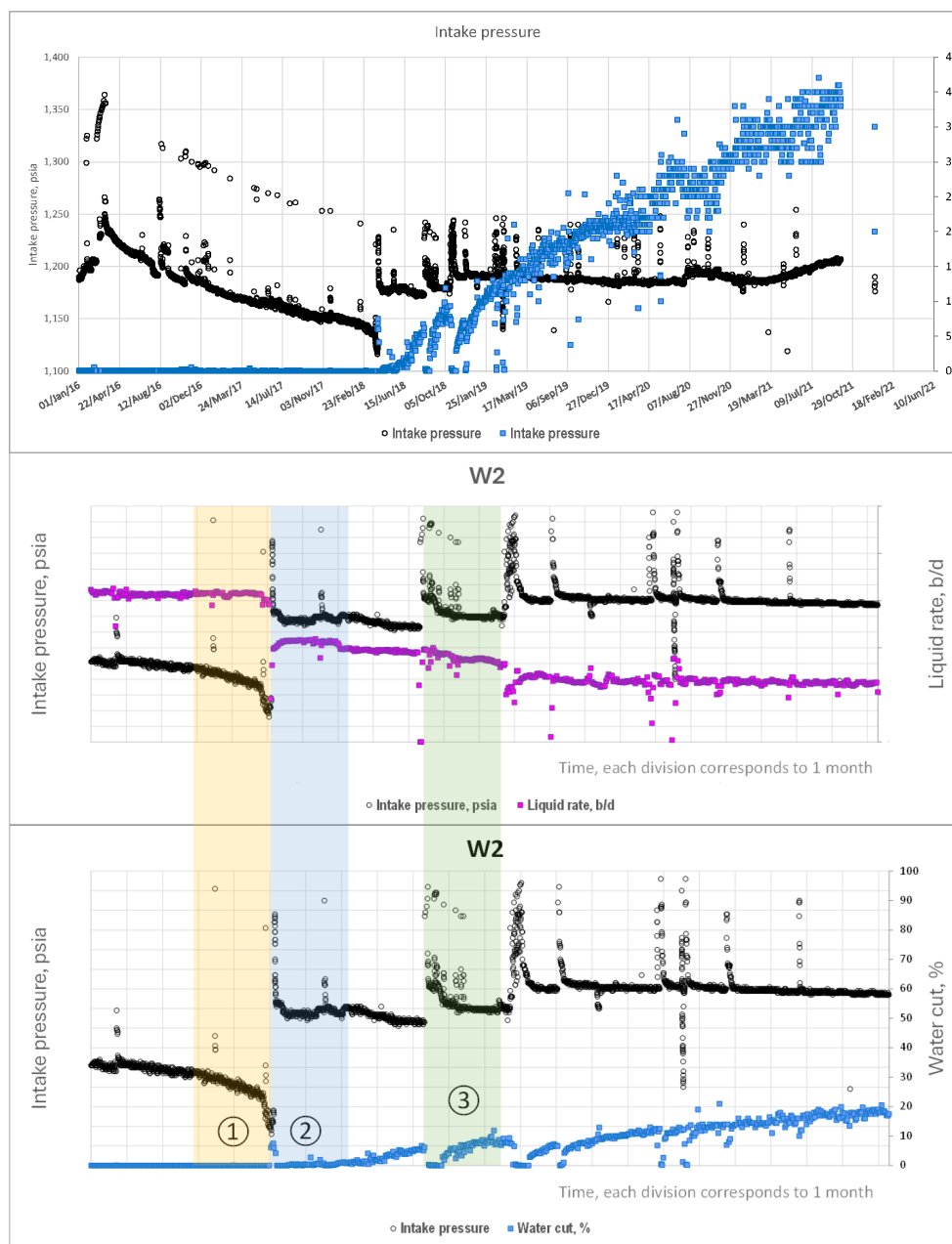


Figure 11—Detailed View of the Bottomhole Pressure, Liquid Rate and Water-Cut Behavior of Well W2

Similarly, as well W1, Panel 1 marks a change in the trend of the well's flowing bottomhole pressure indicating water rising from the bottom aquifer towards the well. Notice the steep decrease of flowing bottomhole pressure as water approaches the wellbore. It is an effective notice of 60 days that water is on its way towards the wellbore. The last 10 days of the period are characterized by a strong decline of flowing bottomhole pressure and at the end of it water-cut jumped from 0% to 7%.

Panel 2 starts with a reduction of liquid rate of 30%. Water disappeared for about 1 week and it came back steadily keeping the liquid rate constant.

Panel 3 illustrates additional attempts to reduce the water-cut by reducing the liquid rate, which became progressively less effective.

Well W5

Well W-5 is located in the disturbed band near the main fracture running SW-NE. Figures 12 and 13 shows the pressure behavior along with its water-cut.

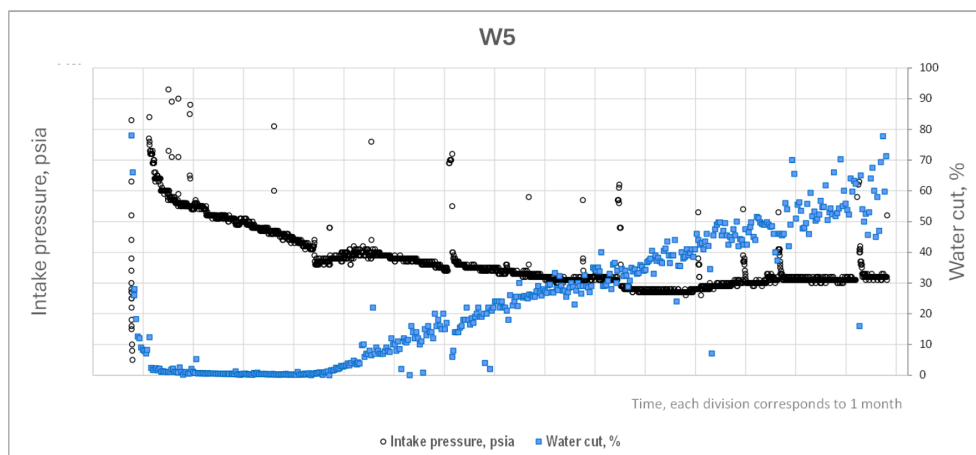


Figure 12—Bottomhole pressure and water-cut of well W5

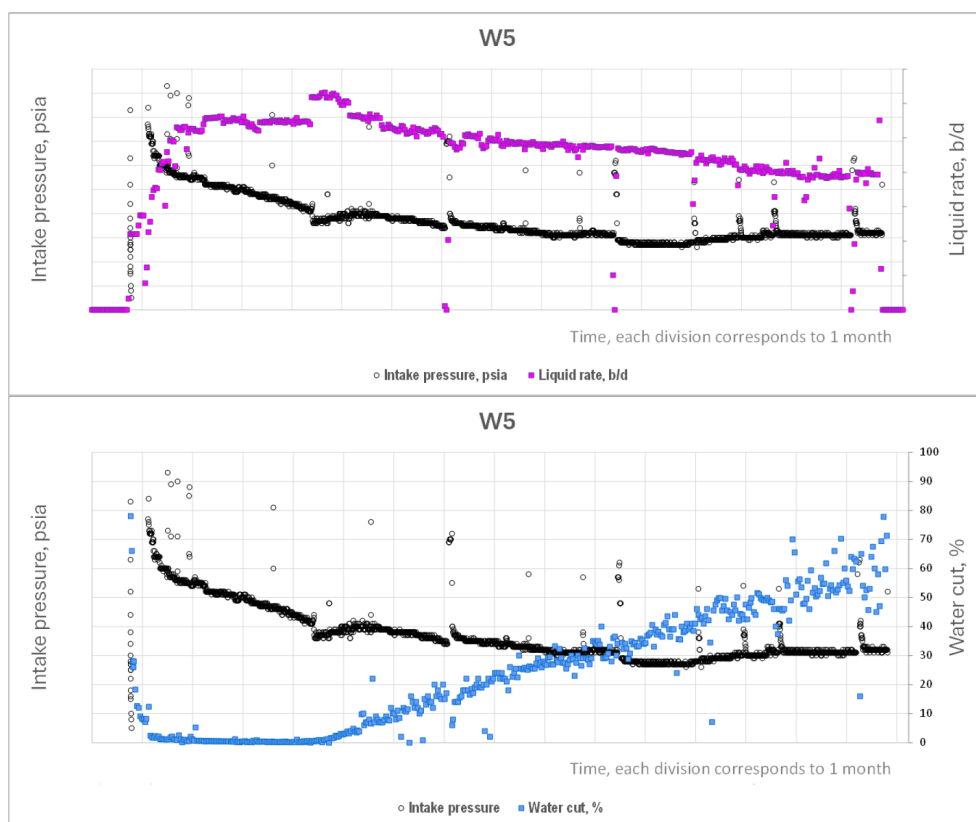


Figure 13—Bottomhole pressure and water-cut of well W5

Well W5 is located within the heavily fractured disturbed zone adjacent to the main SW–NE fault corridor. The fracture density in this area is so high that W5 behaves more like a homogeneous matrix producer than a typical fractured well. Notably, variations in bottomhole pressure have little to no impact on the water cut evolution. This suggests that water production is not governed by gravity segregation, but rather by the presence of numerous high-permeability fracture pathways that allow water to reach the wellbore directly.

Well W4

Well W-4 is located away from the main fractured zones and does not exhibit the typical behavior of wells situated in fractured media (see Figures 14 and 15). While its response bears some resemblance to that of Well W5, the underlying cause is the opposite: in this case, the absence of significant fracturing results in a predominantly matrix-dominated flow regime. As shown in Figure 14, the steady decline in pressure does not appear to influence the water cut, suggesting that the displacement process is mainly governed by relative permeability effects, as is characteristic of matrix flow.

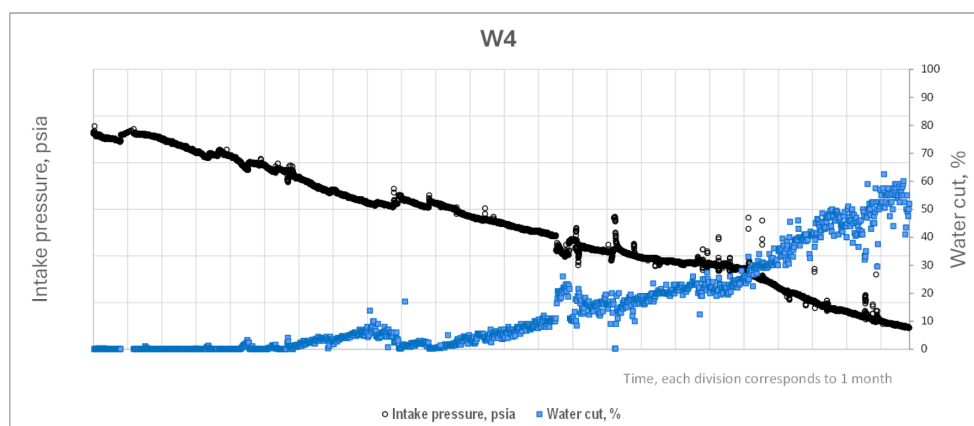


Figure 14—Bottomhole pressure and water-cut of well W4

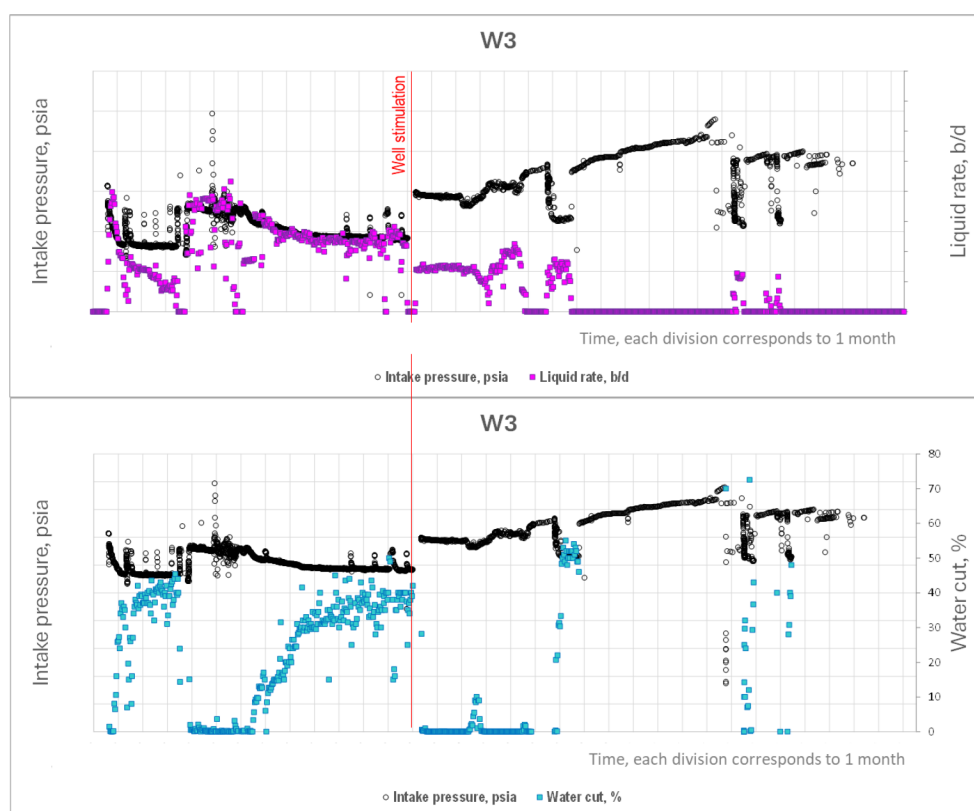


Figure 15—Bottomhole Pressure, Water-Cut and Liquid Rate of Well W3

Well W3

Well W3 is located adjacent to a secondary fault trending east–west. Production began with a low water-cut, which gradually increased to over 30%. The production rate was maintained at that level by managing

the contribution from the open intervals. A Production Logging Tool (PLT) survey was conducted to assess inflow distribution, after which a stimulation treatment was performed with the goal of enhancing the well's productivity index (PI). As shown in Figure 15, the stimulation significantly improved well performance, likely by connecting the wellbore to a more extensive fracture network. The result was a threefold increase in the PI.

Conclusions

A detailed characterization of fault and fracture systems and other associated flow-enhancing geological features by combining wireline logs, fieldwork and field analogues, seismic interpretation and dynamic parameters has proven to be essential for understanding and further predict well behavior patterns in this fractured carbonate field. However, to maintain the effectiveness of this methodology assessment, the multidisciplinary Subsurface team needs to possess the necessary flexibility and adaptability to quickly incorporate new information and conclusions from each evaluation stage in this continuous and retro-fed workflow.

The authors have identified the following main conclusions from this evaluation:

1. Complex reservoir architecture often requires re-learning of reservoir understanding and adjustment of conceptual geological models to the fluid flow dynamics in the reservoir.
2. Continuous dynamic monitoring combined with adaptive representation of flow paths in reservoir models are dependent on detailed fault and fracture network categorization and their spatial dynamic behavior through time.
3. Water production in gravity-controlled systems can be controlled with changes in the drawdown pressure.
4. Selective completions are essential for promoting a uniform inflow profile along the wellbore. Similarly, selective stimulation techniques are critical for achieving a more balanced productivity index (PI) across all perforated intervals.
5. Timely adjustments to drawdown pressure are necessary to control water production effectively. These adjustments should be implemented as soon as changes in the trend of bottomhole flowing pressure are observed. Early intervention through well and reservoir management (WRM) can significantly influence well performance and prolong productive life.
6. Water influx from bottom aquifers requires considerable energy to reach the wellbore. Its approach can be identified by a noticeable decline in flowing bottomhole pressure, which becomes steeper as the waterfront nears the well.
7. The team needs to be able to incorporate learnings and take the necessary actions as fast as possible and sometimes independently from what direct model observations indicate, since models may not be able to capture some fine details or it may not be possible to update them with frequently enough to reflect correctly the current stage in the life of the field.

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